

Phase 8: Project Demonstration — Solo Developer Execution & Ownership Audit

Project Name: Cab Booking (`UCab`)
Project ID: `N/A (Solo Track Submission)`
Team Member / Solo Developer: Shaik Sumiya Zainab

1. Solo Track Ownership Declaration

This project (`UCab`) was designed, engineered, tested, and documented completely by a single developer (`Shaik Sumiya Zainab`). Rather than dividing workload across multiple team members, full-stack accountability was maintained across every layer of the application stack.

2. Complete Lifecycle Ownership Matrix

Project Domain / Discipline Specific Responsibilities Handled Key Deliverables & Artifacts Produced

****1. UI/UX & Design Systems**** Conducted user empathy mapping for riders and dispatchers; designed custom glassmorphism CSS design system (`index.css`) with curated typography and color tokens. `Empathy\_Map.md`, `Customer\_Journey\_Map.md`, `index.css`, `App.css`.

****2. Frontend Engineering (React + Vite)**** Built 11 distinct user and admin page components; constructed role-protected route wrappers (`ProtectedRoutes.jsx`); managed global session via `AuthContext.jsx`. `Home.jsx`, `BookCab.jsx`, `UserHome.jsx`, `ReceiptModal.jsx`, `AdminCars.jsx`, `AuthContext.jsx`.

****3. Backend API Development (Node + Express)**** Architected RESTful endpoints (`/api/users`, `/api/cars`, `/api/bookings`, `/api/admin`); implemented `bcrypt` password hashing and `JWT` Bearer token authentication. `server.js`, `authMiddleware.js`, `userController.js`, `carController.js`, `bookingController.js`.

****4. Database Engineering & ODM**** Designed relational Mongoose schemas (`User`, `Car`, `Booking`, `Admin`); engineered the ****Zero-Config Dual-Engine Storage Adapter (`store.js`)**** for universal grading resilience. `User.js`, `Car.js`, `Booking.js`, `config.js`, `store.js`, `data.json`.

****5. Quality Assurance & Testing**** Executed comprehensive end-to-end testing across login flows, promo code discounting (`UCAB20`), and simulated offline database failover (`data.json`). `Test\_Cases\_and\_Performance.md`, `Performance\_Testing.md`.

****6. Documentation & Template Sync**** Formatted and produced all exhaustive Phase 1 through Phase 8 markdown reports adhering strictly to the ****AI-ML and GEN-AI Track Project Template****. All `.md` deliverable reports inside folders `1.` through `8.`.

3. Solo Presentation & Defense Roles

During evaluation and project viva, the solo developer will execute all demonstration roles seamlessly:

1. Architect Role: Explaining the rationale behind the dual-engine storage adapter (`store.js`) and system DFDs.
2. Rider Role: Demonstrating live cab selection, promo discounting (`UCAB20`), and QR invoice generation (`ReceiptModal.jsx`).
3. Dispatcher Role: Demonstrating real-time fleet inventory CRUD (`AddCar.jsx`) and dispatch status synchronization (`AdminBookings.jsx`).